

Santiago Reyes, 2174589

Ricardo Trevizo, 2173424

Aqeel Somani, 2364462

Initial Thoughts/Exploration

On our initial inspection of the DBLP dataset, we first looked for interesting features that we could analyze and turn into a story. The number of citations, being a measure of how many times a paper has been cited, could demonstrate a paper's popularity or relevance within the CS field. Additionally, we noticed that papers could be linked by joining their IDs with all reference IDs in other papers, forming a grand network of related papers. Similarly, authors likely also serve as a reference, as researchers will tend to write papers about similar topics. We were caught by surprise that no feature indicated the paper's topic. However, the title and abstract are likely to contain the paper's topic, which could be data mined through some NLP clustering or semi-supervised learning efforts. The venue feature provides a natural context for the topics a paper is likely to cover. Using all of that information, we set out a proposal.

Proposal

We want to explore the different topics of Computer Science and measure their popularity. By measuring their popularity, we can then find how the field has changed over time, such as topics that have increased, decreased, and stayed consistent in popularity. We aim to do so by performing tasks that we'll break down into four stages:

Stages

1. Pre-Topic Discovery EDA

From our initial findings, a pre-topic discovery EDA would be best to identify how to "attack" the dataset. We'll start with plots showing some univariate findings, such as the distributions of our features, and we'll see how well "number of citations" maps to our other features. Using our year feature, we'll also investigate each feature and its change over time. We'll aim to remove noisy data, as well.

2. Topic Clustering

We'll perform some clustering on our dataset to datamine our topics. We'll be relying heavily on the title and abstract to provide correlation between papers, and we'll also include authors, as they are likely to write papers with similar topics. References with other papers will also prove useful in this step.

3. Topic Labeling/Classification

Once we are able to define and grab our class labels, we can perform some sort of semi-supervised learning to label the rest of our data. We can then test several classification models to see how well any general model can learn our patterns, using those results to then tune our semi-supervised learning to achieve better results.

4. Post-Topic Discovery EDA

Once we have data mined and successfully assigned labels to each data point, we'll perform EDA using our new class labels. We'll look at feature distributions, how topics have changed over time, and which topics are the most relevant. Additionally, we'll inspect what our models had difficulty classifying and how we can improve.

Challenges:

Natural Language Processing will be the trickiest part of our assignment, as it'll produce high-dimensional data. Additionally, typical methods of NLP, such as a Bag-of-Words, will fail to pick up related terms as they can't extract semantic meaning.

We'll also need to perform some data transformation for our missing/noisy data. We'll likely use some interpolation/statistical methods, but we don't want to misrepresent data or cut out data that could be important.

One issue could stem from our lack of experience in collaborating on Data Science projects. Up until this point, we've only worked on our individual notebooks, and now we'll be working in Google Colab. Alongside making sure we're coordinating on our findings, we want to make sure we aren't overwriting other people's work.

Data Preprocessing Method

For our data preprocessing step, we'll first join our data together into one DataFrame. Then, we'll encode our high-cardinality data, such as the abstract and title, through TD-IDF vectorization, where we'll identify low-importance word features and remove them. We'll aim to fill in missing values, likely through either mean/median substitution or linear interpolation. Then, we perform some feature engineering, where we can detect if any features can be made from the original features. Finally, we'll normalize our data using Z-score normalization.

Ideas behind each method

The analysis of the dataset will be conducted through three different methods: EDA, Clustering, and Classification. EDA serves as the foundation by developing the structure of the data. It will serve to organize the features that are most important to the analysis, such as frequency of citations and publication years, to inform the next steps of modeling. Clustering will be applied to the papers to group them into various topics based on the title and abstract. Algorithms like K-Means and TF-IDF will be used to extract similar words and minimize commonly used phrases and words. Lastly, classification will be used to assign different clusters to their levels of influence, allowing the model to learn what features produce results and then have the ability to generalize this onto new data. Together, these methods work to organize the dataset into its respective topics and quantify their level of influence.

Usefulness of analysis

Through this analysis, we can understand how research has evolved throughout the history of Computer Science. Citation analysis has always been a reliable way to evaluate the significance of contributions in this field. These highly cited works are often the basis upon which entire new subfields can be developed and researched. This analysis, by identifying the most influential papers and the respective clusters of topics, provides a structured format of how this field has evolved and shifted throughout time. This makes it easy to visualize which directions of research contributed to the most meaningful advancements. Specifically in the DBLP dataset, where there is a large volume of publications, using data-driven machine learning can reveal patterns that were not obvious before. Overall, the insights of this analysis can help future researchers identify high-impact topics from history as well as provide insight into topics that may not have been explored fully.

Roles:

- Ricardo: Clustering
 - Ordering by certain features, K-Means, Hierarchical methods (Dendrogram), Density-based clustering w/ DBSCAN
 - Creating class labels using natural forming clusters
 - Detect any particular anomalies in the dataset
 - Identify what makes a paper popular
- Aqeel: EDA
 - Explore variance across all features
 - Analyze linear relationships between features (author x venue)
 - Outlier/Noise Reduction
 - Univariate Analysis: Box plots, scatter plots, PCAs, feature importance, histograms
 - Identify patterns using data visualization
 - Testing basic assumptions
 - Do older papers tend to have more citations?
 - Do papers with more citations have different venues than papers with less citations?
- Santiago: Regular Classification
 - Use topic labels produced by clustering as class labels, then train models to learn whether title and abstract features can reliably predict a paper's topic groups
 - Use Decision Trees, Random Forest, and Multinomial Naive Bayes
 - Evaluate metrics using Macro-F1, Micro-F1, Accuracy, Precision, Recall, and training time
 - Testing assumptions:
 - Adding venue information to title and abstract features will improve topic classification performance
 - Topic clusters with more distinct vocabulary will be easier to classify compared to others
 - Some topic clusters will have a higher average number of citations which suggest what areas have had the greatest influence